

ASIM ASLAH P M

AI/ML ENGINEER | SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Computer Science graduate pursuing an M.Tech in Artificial Intelligence and Data Science, with project experience in computer vision, privacy-preserving image and video processing, image-to-3D reconstruction, time-series forecasting, and secure software systems. Builds Python applications by integrating YOLO, SAM, TripoSR, OpenCV, and Gradio; targeting entry-level AI/ML and software engineering roles.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

AI/ML & Computer Vision: Machine Learning, Computer Vision, Image Processing, Object Detection, YOLO, Segment Anything Model (SAM), OpenCV, TripoSR

Data & Visualization: NumPy, Pandas, Matplotlib, Time-Series Forecasting

Web & Platforms: Django, Gradio, Supabase

Developer Tools: Git, GitHub, VS Code, Overleaf

Foundations: Data Structures, Algorithms, Object-Oriented Programming, Cryptography

SELECTED PROJECTS

Privacy-Preserving Image & Video Anonymization | Jun 2026 - Present

Technologies: Python, YOLO, Segment Anything Model (SAM)

- Engineered a Python pipeline to detect and anonymize sensitive regions in image and video inputs.
- Integrated YOLO object detection with SAM segmentation to generate object-level masks for automated anonymization.
- Applied masking and image-processing operations to detected regions across images and video frames.

Image-to-3D Generation Application | Jan 2026

Technologies: Python, TripoSR, Gradio

- Built an interactive Gradio application that converts a single 2D image into a 3D asset using TripoSR.
- Implemented model inference and visualization/export workflows for GLB and OBJ formats, with CPU-compatible execution.

Security and Forecasting Energy Distribution System | May 2024

Technologies: Python, AES-256 Cryptography, Time-Series Forecasting

- Developed a Python prototype combining AES-256 encryption with time-series forecasting to protect consumer energy data and estimate short- and long-term demand.
- Integrated cryptography and predictive analytics into a unified workflow designed to support demand planning and data-driven power distribution decisions.

AI Virtual Keyboard | Aug 2023

Technologies: Python, OpenCV

- Built an OpenCV-based virtual keyboard using live camera input and hand tracking for real-time gesture-driven text entry.
- Implemented finger-gesture recognition and key-selection logic to enable touch-free human-computer interaction.

EDUCATION

Master of Technology in Artificial Intelligence and Data Science | Aug 2025 - Present

Institution: Government Engineering College Palakkad

Bachelor of Technology in Computer Science | Jan 2020 - Jan 2024

Institution: MES College of Engineering